



PRODUCT NAME

Professional Roof Cleaner Formula

Description

A cost-effective, non-toxic liquid formulation expertly crafted to combat algae and mold on roofing surfaces. This formula is designed to be non-corrosive to roofing materials and safe for plants and landscaping, providing an environmentally friendly cleaning solution.

Technical Specifications

pH Level:	7
Viscosity:	Low
Shelf Life:	24 months
Batch Size:	50-200 liters
Processing Time:	Approximately 2 hours
Temperature:	Room temperature (20-25°C)
Storage Conditions:	Store in a cool, dry place away from direct sunlight.
Equipment:	Mixing vessel, stirring apparatus, pH meter, filtration system, packaging equipment
Certification:	Compliant with EPA regulations for non-toxic cleaning agents

Formulation Table

Sr.No	Ingredient	INCI Name	%	Function
1	Water	Aqua	75.0%	Solvent and base for the formulation
2	Sodium Bicarbonate	Sodium Bicarbonate	10.0%	Alkaline agent effective against mold and algae
3	Ethoxylated Alcohol	Alcohols, C10-16, ethoxylated	7.0%	Surfactant for cleaning and removing organic residues
4	Citric Acid	Citric Acid	4.0%	pH adjuster to maintain exact pH of 7
5	Benzalkonium Chloride	Benzalkonium Chloride	2.5%	Algaecide and fungicide to effectively combat algae and mold
6	Sodium Gluconate	Sodium Gluconate	1.0%	Corrosion inhibitor to protect roofing materials
7	Essential Oil Blend	Various Fragrance Oils	0.5%	To impart a mild, pleasant scent
Total			100%	

Manufacturing Process

Phase 1: Phase 1: Preparation

1. Prepare all necessary ingredients and equipment.
2. Ensure the mixing vessel is clean and sanitized.

Phase 2: Phase 2: Mixing

1. Add Water to the mixing vessel.
2. Slowly add Sodium Bicarbonate while continuously stirring until dissolved.
3. Incorporate Ethoxylated Alcohol and mix thoroughly.
4. Add Benzalkonium Chloride and continue mixing.

5. Adjust pH by adding Citric Acid until pH is exactly 7.
6. Add Sodium Gluconate while stirring gently.
7. Incorporate Essential Oil Blend for fragrance.

Phase 3: Phase 3: Finalization

1. Ensure the solution is homogenous.
2. Filter the final product to remove any impurities.
3. Package the product in appropriate, labeled containers.

Required Equipment

Mixing vessel, stirring apparatus, pH meter, filtration system, packaging equipment

Safety Precautions

Handling: Wear appropriate PPE including gloves, safety glasses, lab coat.

PPE Requirements: Chemical-resistant gloves, safety goggles, protective clothing.

Storage: Store in cool, dry place away from direct sunlight. Keep containers tightly closed.

Storage Conditions: Store in a cool, dry place away from direct sunlight.

Packaging Notes

Packaging: Use chemically compatible containers (HDPE, glass, or PET).

Labeling: Include product name, ingredients, usage instructions, and safety warnings.

Certification: Compliant with EPA regulations for non-toxic cleaning agents

Scaling Note

Lab Scale: This formulation is designed for laboratory testing and development.

Pilot Scale: For pilot production, scale proportionally and verify all parameters.

Production Scale: Consider equipment limitations, mixing efficiency, and process validation.

Batch Size: Current formulation is optimized for 50-200 liters.

Scaling Factor: Maintain ingredient ratios while adjusting processing parameters as needed.

Product Usage Instructions & Application Guidelines

Apply the product evenly to the roof surface using a sprayer. Allow the solution to sit for 15-30 minutes to effectively break down algae and mold. Rinse thoroughly with water. Use in well-ventilated areas and avoid contact with skin and eyes.